

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Audio Processor
Manufacturer: BSS
Firmware Version: 86.2.2
6.00.R5 (London Architect)
1.96.0 (Audio Architect)
Model(s): BLU-102, BLU-800, BLU-806, BLU-805, BLU-326, BLU-325, BLU-320, BLU-160, BLU-120,
BLU-101, BLU-100, BLU-GPX, BLU-50, BLU-103

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.2.0

Version History

Module Version	Date	Notes
1_6_1_0	3/5/2019	Updated HiQAddress command parameters to be non-case-sensitive. Added Clear Redial value to Hook command. Updated Hook and VoIP Hook functionality.
1_5_4_0	7/16/2018	Added note regarding use of Gain command.
1_5_3_1	2/9/2018	Updated Module notes.

Global Scripter Module Communication Sheet

1_5_3_0	9/8/2017	Added VoipCallerID. Updated Doc to Rev. B
1_5_1_0	3/24/2017	Updated logic for setting HiQnet Node Address to account for Audio Architect software.
1_5_0_1	2/8/2017	Updated BSS software notes.
1_5_0_0	1/17/2017	Added VoIP commands for model BLU-103. Updated Logic Source Select and Source max value. Removed non-VoIP commands from Blu-103 model. Removed Hook command from BLU-80 model (not supported).
1_1_4_0	6/30/2016	Module standards update.
1_1_3_0	3/28/2016	Initial Version.

Module Notes

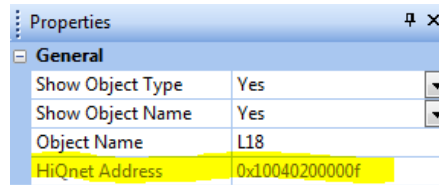
- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- Status will only update when a change is made to the device other than through the module (the BSS software, e.g.).
- The device can be configured using either the London Architect or the Audio Architect software.

Supported Classes and Examples

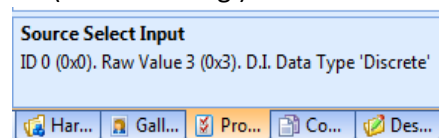
SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='BLU-102')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='BLU-102')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 1023, Model='BLU-102')</code>

London Architect Information:

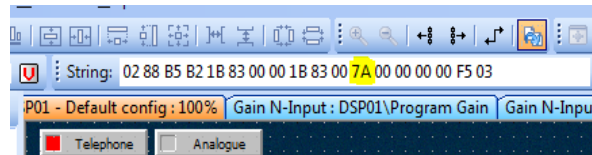
All commands require a HiQAddress parameter, which can be obtained from object Properties tab in the General section, as shown below. **The HiQAddress is case sensitive and must be entered in lowercase alpha numeric characters** ('0x10040200000f', e.g.).



Some commands require an ID parameter, which can be obtained by selecting the parameter in the Parameters section of the Properties Tab. The ID will be displayed at the bottom of the Properties tab, as shown below. **The ID will need to be entered in integer format** (0x0A or 10 e.g.)

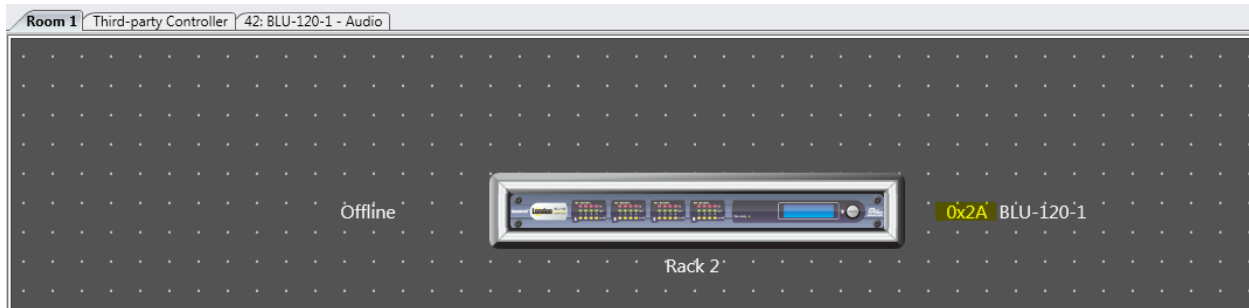


For Call Status, the ID will be displayed in the String field when the call status field is selected. The highlighted byte shown below corresponds to the ID used when programming the ID value into Global Scriptor® (122 for 0x7A, e.g.).

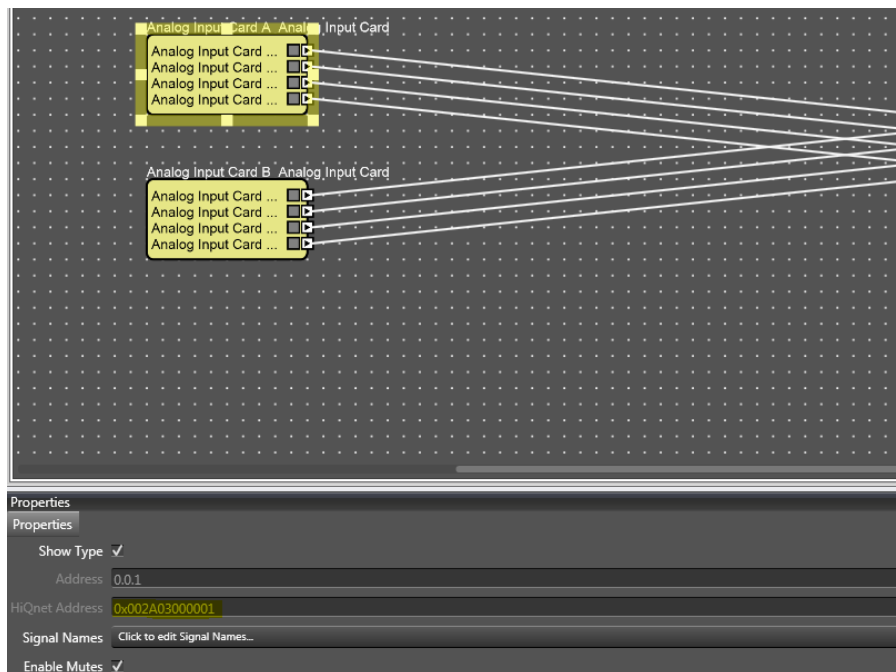


Audio Architect Information:

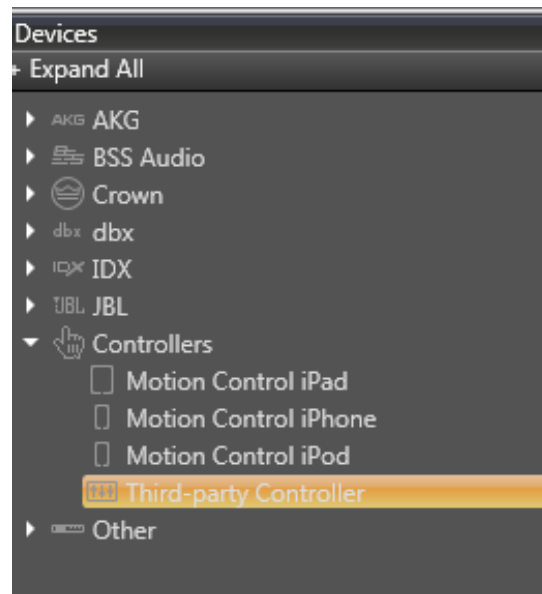
HiQnet Node Address must be programmed for the module to work. It can be obtained by selecting a Room from the Venue, and will be shown next to the device name. The value is highlighted below and should be programmed in the following format: InterfaceName.HiQnetNodeAddress = '0x002A', e.g.



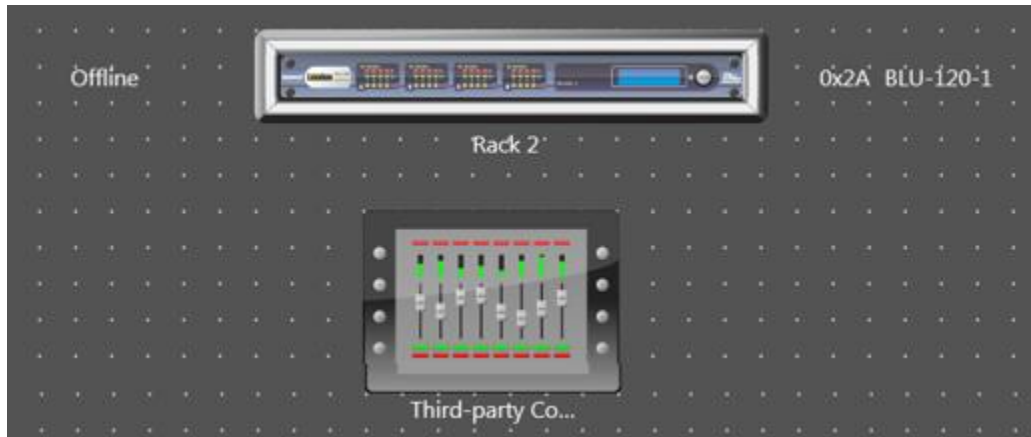
Most commands require a HiQAddress parameter, which can be obtained by first double clicking on a desired BSS device within a room. Upon doing so, single click on the desired block to control and the value can be seen in the Properties tab, as shown in the picture highlighted below. **The HiQAddress is case sensitive and must be entered in lowercase alpha numeric characters ('0x002a03000001', e.g.).**



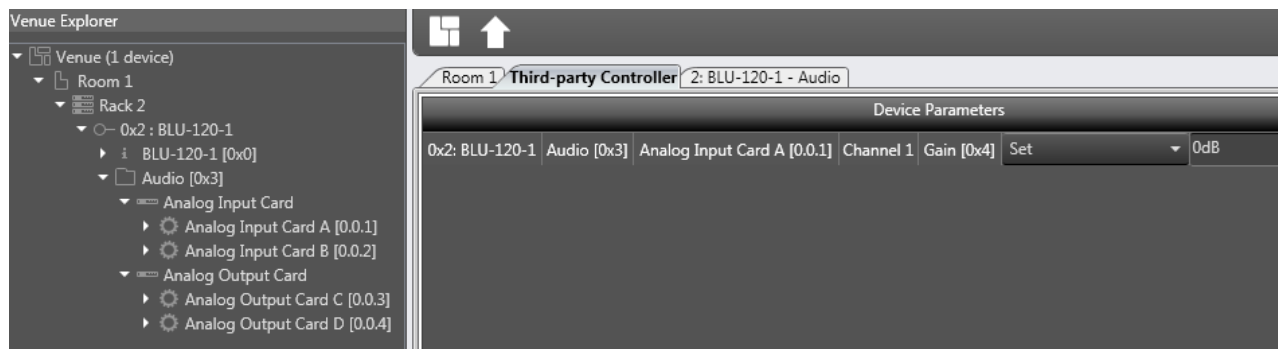
Some commands require an ID parameter, which can be obtained by first adding a Third-party Controller. To do this, select the Controllers drop down under the Devices panel on the left, and drag and drop in Third-party Controller into the room.



By double clicking on the Third-party Controller, a listed of available commands for third-party control can be seen for the devices that are in that room.



For example, below is a BLU-120 configuration, and available commands can be seen in the expanded menu on the left. Drag and drop the desired commands into the Device Parameters panel to the right.



The ID will be displayed next to command name, as shown below. When entering the ID , the ID will need to be entered in integer format (4 for 0x04 or 2000 for 0x7D0, e.g.).

Room 1 Third-party Controller 42: BLU-120-1 - Audio			
0x2A: BLU-120-1	Audio [0x3]	Analog Input Card A [0.0.1]	Channel 1
Gain [0x4]		Set	0dB
Mute [0x7D0]		Set	Unmuted

The VoIP commands require a Line parameter to be specified, which is determined based on which VoIP Line (1 or 2) is needed to control. **VoIP commands are only supported for the BLU-103 model.**

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoAnswer ¹	Value 'Off' '3 Rings' '6 Rings' '9 Rings'	Value '1 Ring' '4 Rings' '7 Rings' '10 Rings'	Value '2 Rings' '5 Rings' '8 Rings'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
# AutoAnswer example InterfaceName.Set('AutoAnswer', 'Off', {'HiQAddress': 'String'})			
Command DTMF ¹	Value '0' '3' '6' '9' , 'Delete'	Value '1' '4' '7' '#' '+'	Value '2' '5' '8' '*' 'Clear'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
# DTMF example InterfaceName.Set('DTMF', '0', {'HiQAddress': 'String'})			
Command Gain ^{2, 3}	Value 0 to 100 in steps of 1		
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'ID'	Qualifier Value 0 to 65535 in steps of 1		
# Gain example InterfaceName.Set('Gain', 100, {'HiQAddress': 'String', 'ID': 65535})			
Command Hook ¹	Value 'On' 'Flash'	Value 'Off' ⁴ 'Clear Redial'	Value 'Redial'
Qualifier Key	Qualifier Value		

'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'Number'	'String'	
# Hook example # Hook Off, ready to dial InterfaceName.Set('Hook', 'Off', {'HiQAddress': 'String'}) # Hook Off and dial the number passed in qualifier InterfaceName.Set('Hook', 'Off', {'HiQAddress': 'String', 'Number': '123456789'})		
Command	Value	
LogicSource	'0' – '1'	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# LogicSource example InterfaceName.Set('LogicSource', '0', {'HiQAddress': 'String', 'ID': 65535})		
Command	Value	
LogicSourceSelect	0 – 32	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# LogicSourceSelect example InterfaceName.Set('LogicSourceSelect', 32, {'HiQAddress': 'String', 'ID': 65535})		
Command	Value	
LogicValue	'String'	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# LogicValue example InterfaceName.Set('LogicValue', 'String', {'HiQAddress': 'String', 'ID': 65535})		
Command	Value	Value
Mute ²	'On'	'Off'
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# Mute example InterfaceName.Set('Mute', 'On', {'HiQAddress': 'String', 'ID': 65535})		
Command	Value	
Preset	0 – 65535	

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Qualifier Key 'Type'	Qualifier Value 'Venue'	Qualifier Value 'Param'	
# Preset example InterfaceName.Set('Preset', 65535, {'Type': 'Venue'})			
Command Source ²	Value 0 – 128		
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'ID'	Qualifier Value 0 to 65535 in steps of 1		
# Source example InterfaceName.Set('Source', 128, {'HiQAddress': 'String', 'ID': 65535})			
Command VoIPAutoAnswer	Value 'On'	Value 'Off'	
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'Line'	Qualifier Value '1' – '2'		
# VoIPAutoAnswer example InterfaceName.Set('VoIPAutoAnswer', 'On', {'HiQAddress': 'String', 'Line': '1'})			
Command VoIPDTMF	Value '0' '3' '6' '9' , '1' '4' '7' '#' 'Clear'	Value '2' '5' '8' '*' 'Delete'	
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'Line'	Qualifier Value '1' – '2'		
# VoIPDTMF example InterfaceName.Set('VoIPDTMF', '0', {'HiQAddress': 'String', 'Line': '1'})			
Command VoIPHook	Value 'Pick Up/Hang Up' 'Dial'	Value 'Hold' 'Redial'	Value 'Reject' 'Clear Redial'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'Line'	Qualifier Value '1' – '2'		
Qualifier Key	Qualifier Value		

'Number' ³	'String'
<pre># VoIPHook example InterfaceName.Set('VoIPHook', 'Pick Up/Hang Up', {'HiQAddress': 'String', 'Line': '1'}) InterfaceName.Set('VoIPHook', 'Dial', {'HiQAddress': 'String', 'Line': '1', 'Number' : '192.168.254.254'})</pre>	

¹ Not supported for BLU-GPX, BLU-16, BLU-32, BLU-50, BLU-80, BLU-100, BLU-101, or BLU-103 models.

² Not supported for BLU-GPX model.

³ Numer qualifier is needed only when the value is 'Dial'

⁴ The qualifier Number is optional, when passed the module will dial the number, if empty or None BSS will only go Off hook (open line)

⁹ For this command to work properly, the type must be set to Set Percent (for control) and Subscribe Percent (for status) under the Third-party Controller tab within the BSS software.

Status Available

For all commands excluding VoIPCallerID, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AutoAnswer ¹	Value 'Off' '3 Rings' '6 Rings' '9 Rings'	Value '1 Ring' '4 Rings' '7 Rings' '10 Rings'	Value '2 Rings' '5 Rings' '8 Rings'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
<pre># AutoAnswer examples InterfaceName.Update('AutoAnswer', {'HiQAddress': 'String'}) Value = InterfaceName.ReadStatus('AutoAnswer', {'HiQAddress': 'String'}) InterfaceName.SubscribeStatus('AutoAnswer', None, FeedbackHandler)</pre>			
Command CallStatus ¹	Value 'Incoming Call'	Value 'Idle'	
Qualifier Key 'HiQAddress'	Qualifier Value 'String'		
Qualifier Key 'ID'	Qualifier Value 0 to 65535 in steps of 1		
<pre># CallStatus examples InterfaceName.Update('CallStatus', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('CallStatus', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('CallStatus', None, FeedbackHandler)</pre>			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
<pre># ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)</pre>			

Command	Value	
Gain ^{2, 3}	0 to 100 in steps of 1	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# Gain examples InterfaceName.Update('Gain', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('Gain', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('Gain', None, FeedbackHandler)		
Command	Value	Value
Hook ¹	'On'	'Off'
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
# Hook examples InterfaceName.Update('Hook', {'HiQAddress': 'String'}) Value = InterfaceName.ReadStatus('Hook', {'HiQAddress': 'String'}) InterfaceName.SubscribeStatus('Hook', None, FeedbackHandler)		
Command	Value	
LogicSource	'0' – '1'	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# LogicSource examples InterfaceName.Update('LogicSource', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('LogicSource', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('LogicSource', None, FeedbackHandler)		
Command	Value	
LogicSourceSelect	0 – 32	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	
'ID'	0 to 65535 in steps of 1	
# LogicSourceSelect examples InterfaceName.Update('LogicSourceSelect', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('LogicSourceSelect', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('LogicSourceSelect', None, FeedbackHandler)		
Command	Value	
LogicValue	'String'	
Qualifier Key	Qualifier Value	
'HiQAddress'	'String'	
Qualifier Key	Qualifier Value	

'ID'	0 to 65535 in steps of 1	
# LogicValue examples InterfaceName.Update('LogicValue', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('LogicValue', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('LogicValue', None, FeedbackHandler)		
Command Mute ²	Value 'On'	Value 'Off'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'ID'	Qualifier Value 0 to 65535 in steps of 1	
# Mute examples InterfaceName.Update('Mute', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('Mute', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)		
Command Source ²	Value 0 – 128	
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'ID'	Qualifier Value 0 to 65535 in steps of 1	
# Source examples InterfaceName.Update('Source', {'HiQAddress': 'String', 'ID': 65535}) Value = InterfaceName.ReadStatus('Source', {'HiQAddress': 'String', 'ID': 65535}) InterfaceName.SubscribeStatus('Source', None, FeedbackHandler)		
Command VoIPAutoAnswer	Value 'On'	Value 'Off'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'Line'	Qualifier Value '1' – '2'	
# VoIPAutoAnswer examples InterfaceName.Update('VoIPAutoAnswer', {'HiQAddress': 'String', 'Line': '1'}) Value = InterfaceName.ReadStatus('VoIPAutoAnswer', {'HiQAddress': 'String', 'Line': '1'}) InterfaceName.SubscribeStatus('VoIPAutoAnswer', None, FeedbackHandler)		
Command VoIPCallerID	Value 'String'	
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'Line'	Qualifier Value '1' – '2'	
# VoIPCallerID examples Value = InterfaceName.ReadStatus('VoIPCallerID', {'HiQAddress': 'String', 'Line': '1'})		

InterfaceName.SubscribeStatus('VoIPCallerID', None, FeedbackHandler)		
Command VoIPHoldStatus	Value 'On Hold'	Value 'Not on Hold'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'Line'	Qualifier Value '1' – '2'	
# VoIPHoldStatus examples InterfaceName.Update('VoIPHoldStatus', {'HiQAddress': 'String', 'Line': '1'}) Value = InterfaceName.ReadStatus('VoIPHoldStatus', {'HiQAddress': 'String', 'Line': '1'}) InterfaceName.SubscribeStatus('VoIPHoldStatus', None, FeedbackHandler)		
Command VoIPHookStatus	Value 'On'	Value 'Off'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'Line'	Qualifier Value '1' – '2'	
# VoIPHookStatus examples InterfaceName.Update('VoIPHookStatus', {'HiQAddress': 'String', 'Line': '1'}) Value = InterfaceName.ReadStatus('VoIPHookStatus', {'HiQAddress': 'String', 'Line': '1'}) InterfaceName.SubscribeStatus('VoIPHookStatus', None, FeedbackHandler)		
Command VoIPRingingStatus	Value 'Ringing'	Value 'Not Ringing'
Qualifier Key 'HiQAddress'	Qualifier Value 'String'	
Qualifier Key 'Line'	Qualifier Value '1' – '2'	
# VoIPRingingStatus examples InterfaceName.Update('VoIPRingingStatus', {'HiQAddress': 'String', 'Line': '1'}) Value = InterfaceName.ReadStatus('VoIPRingingStatus', {'HiQAddress': 'String', 'Line': '1'}) InterfaceName.SubscribeStatus('VoIPRingingStatus', None, FeedbackHandler)		

¹ Not supported for BLU-GPX, BLU-16, BLU-32, BLU-50, BLU-80, BLU-100, BLU-101, or BLU-103 models.

² Not supported for BLU-GPX model.

³ For this command to work properly, the type must be set to Set Percent (for control) and Subscribe Percent (for status) under the Third-party Controller tab within the BSS software.

Cable and Adapter Requirements

Captive screw to Female DB9 RS232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

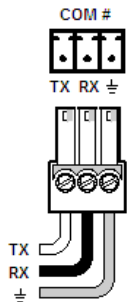
Data Bits: 8

Parity: None

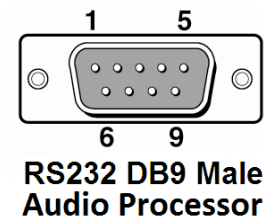
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	1023
Logon Credentials Supported:	No
Multi-Connection	Yes
Capabilities:	
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device